1.44 Hidden Polynomials

Worked Solutions

1 a)

$$5x^4 + 2x^2 - 3 = (x^2 + 1)(5x^2 - 3) = 0$$
$$\therefore x^2 = -1, x^2 = \frac{3}{5}$$

$x^2 = -1$ has no solution.

$$x = \pm\sqrt{\frac{3}{5}}$$

1 b)

$$3x - 16\sqrt{x} - 35 = (\sqrt{x} - 7)(3\sqrt{x} + 5) = 0$$
$$\therefore \sqrt{x} = -\frac{5}{3}, \sqrt{x} = 7$$

$\sqrt{x} = -\frac{5}{3}$ has no solution.

$$x = 49$$

1 c)

$$2x^{\frac{2}{3}} + 7x^{\frac{1}{3}} - 15 = 0$$
$$\left(2x^{\frac{1}{3}} - 3\right)\left(x^{\frac{1}{3}} + 5\right) = 0$$
$$\therefore x^{\frac{1}{3}} = \frac{3}{2}, x^{\frac{1}{3}} = -5$$

$x^{\frac{1}{3}} = -5$ has no solution.

$x^{\frac{1}{3}} = \frac{3}{2}$ so $x = \frac{27}{8}$

1 d)

$$\frac{6}{x^2} + \frac{7}{x} = 5$$
$$6 + 7x = 5x^2$$
$$5x^2 - 7x - 6 = 0$$
$$(x - 2)(5x + 3) = 0$$
$$\therefore x = 2, x = -\frac{3}{5}$$

1 e)

$$7^x + 7^{2x} - 2 = 0$$
$$(7^x + 2)(7^x - 1) = 0$$
$$\therefore 7^x = -2, 7^x = 1$$

$7^x = -2$ has no solution.

$$x = 0$$

1 f)

$$\begin{aligned} 2^{2x} - 3 \times 2^{-1} \times 2^x + 2^{-1} &= 0 \\ 2^{2x} - \frac{3}{2} \times 2^x + \frac{1}{2} &= 0 \\ 2 \times 2^{2x} - 3 \times 2^x + 1 &= 0 \\ (2^x - 1)(2 \times 2^x - 1) &= 0 \\ \therefore 2^x = 1, 2^x &= \frac{1}{2} \\ \therefore x = 0, x &= -1 \end{aligned}$$

1 g)

$$\begin{aligned} 3^{3x+1} + 20 \times 3^{2x} - 7 \times 3^x &= 0 \\ 3 \times 3^{3x} + 20 \times 3^{2x} - 7 \times 3^x &= 0 \\ 3^x(3 \times 3^{2x} + 20 \times 3^x - 7) &= 0 \\ 3^x(3 \times 3^x - 1)(3^x + 7) &= 0 \\ \therefore 3^x = 0, 3^x = -7, 3^x &= \frac{1}{3} \end{aligned}$$

$3^x = 0$ has no solution

$3^x = -7$ has no solution

$x = -1$ is the only solution.

1 h)

$$\begin{aligned} 2^{x+1} + 11 &= \frac{6}{2^x} \\ 2^{2x+1} + 11 \times 2^x &= 6 \\ 2 \times 2^{2x} + 11 \times 2^x - 6 &= 0 \\ (2^x + 6)(2 \times 2^x - 1) &= 0 \\ \therefore 2^x = -6, 2^x &= \frac{1}{2} \end{aligned}$$

$2^x = -6$ has no solution

$2^x = \frac{1}{2} \Rightarrow x = -1$

2)

$$\begin{aligned} 5^8 - 3^8 &= (5^4 - 3^4)(5^4 + 3^4) \\ &= (625 - 81)(625 + 81) \\ &= 544 \times 706 \\ &= 2^5 \times 17 \times 2 \times 353 \\ &= 2^6 \times 17 \times 353 \end{aligned}$$

3)

$$\begin{aligned}
\frac{2x^4+2}{5} &= x^2 \\
2x^4 - 5x^2 + 2 &= 0 \\
(2x^2-1)(x^2-2) &= 0 \\
x^2 = \frac{1}{2} \text{ or } x^2 &= 2 \\
x = \pm\frac{1}{\sqrt{2}} \text{ or } x &= \pm\sqrt{2}
\end{aligned}$$

4)

$$\begin{aligned}
8x^3 + \frac{1}{x^3} &= -9 \\
8x^6 + 1 &= -9x^3 \\
8x^6 + 9x^3 + 1 &= 0 \\
(8x^3+1)(x^3+1) &= 0 \\
\therefore x^3 = -\frac{1}{8}, x^3 &= -1 \\
x = -\frac{1}{2}, x &= -1
\end{aligned}$$

5)

$$\begin{aligned}
2x^6 + \frac{1}{x^3} &= x^3 + 2 \\
2x^9 + 1 &= x^6 + 2x^3 \\
2x^9 - x^6 - 2x^3 + 1 &= 0 \\
(2x^3-1)(x^3+1)(x^3-1) &= 0 \\
\therefore x^3 = \frac{1}{2}, x^3 = -1, x^3 &= 1 \\
x = \frac{1}{\sqrt[3]{2}}, x = -1, x &= 1
\end{aligned}$$

6)

$$\begin{aligned}
\left(x - \frac{4}{x}\right)^2 + 8 &= 6\left(x - \frac{4}{x}\right) \\
y^2 - 6y + 8 &= 0 \\
(y-4)(y-2) &= 0 \\
y = 4 \text{ or } y &= 2 \\
x - \frac{4}{x} = 4 \text{ or } x - \frac{4}{x} &= 2 \\
x^2 - 4 = 4x \text{ or } x^2 - 4 &= 2x
\end{aligned}$$

$$x^2 - 4x - 4 = 0 \text{ or } x^2 - 2x - 4 = 0$$
$$x = 2 \pm \sqrt{2} \text{ or } x = 1 \pm \sqrt{5}$$

7)

$$5x^2 - 12x\sqrt{x} - 39x + 22\sqrt{x} + 24 = 0$$
$$5(\sqrt{x})^4 - 12(\sqrt{x})^3 - 39(\sqrt{x})^2 + 22\sqrt{x} + 24 = 0$$
$$(\sqrt{x} - 4)(5\sqrt{x} + 3)(\sqrt{x} + 2)(\sqrt{x} - 1) = 0$$
$$\therefore \sqrt{x} = -2, \sqrt{x} = -\frac{3}{5}, \sqrt{x} = 1, \sqrt{x} = 4$$
$$\therefore x = 1, x = 16$$

8)

$$3^{2x+1} + 20 \times 3^x + \frac{75}{3^{2x}} = 22 + \frac{220}{3^x}$$
$$3^{4x+1} + 20 \times 3^{3x} + 75 = 22 \times 3^{2x} + 220 \times 3^x$$
$$3 \times 3^{4x} + 20 \times 3^{3x} + 75 = 22 \times 3^{2x} + 220 \times 3^x$$
$$3(3^x)^4 + 20(3^x)^3 + 75 = 22(3^x)^2 + 220(3^x)$$
$$3(3^x)^4 + 20(3^x)^3 - 22(3^x)^2 - 220(3^x) + 75 = 0$$
$$(3^x + 5)^2(3^x - 3)(3 \times 3^x - 1) = 0$$
$$\therefore 3^x = \frac{1}{3}, 3^x = 3, 3^x = -5$$

$3^x = -5$ has no solution

$\therefore x = -1, x = 1$

9 a)

$$3 \times 2^{2x} + 7 \times 2^x - 6 = 0$$
$$(x + 2^x)(3 \times 2^x - 2) = 0$$
$$\therefore 2^x = -3, 2^x = \frac{2}{3}$$

$2^x = -3$ has no solution

$$x = \log_2 \frac{2}{3}$$

9 b)

$$2e^{3x} + 3e^{2x} - 17e^x + 12$$
$$= (e^x + 4)(2e^x - 3)(e^x - 1) = 0$$
$$\therefore e^x = -4, e^x = 1, e^x = \frac{3}{2}$$

$e^x = -4$ has no solution

$x = \ln 1 = 0$ or $x = \ln \frac{3}{2}$

9 c)

$$\begin{aligned} 4 \times 3^x + 13 &= \frac{2}{3^x} + \frac{15}{3^{2x}} \\ 4 \times 3^{4x} + 13 \times 3^{3x} &= 2 \times 3^{2x} + 15 \times 3^x \\ 4 \times 3^{4x} + 13 \times 3^{3x} - 2 \times 3^{2x} - 15 \times 3^x &= 0 \\ 3^x(3^x - 1)(3^x + 3)(4 \times 3^x + 5) &= 0 \\ \therefore 3^x = -3, 3^x = -\frac{5}{4}, 3^x &= 0, 3^x = 1 \end{aligned}$$

$3^x = -3, 3^x = -\frac{5}{4}, 3^x = 0$ have no solution

$3^x = 1 \Rightarrow x = 0$

10 a)

$$\begin{aligned} 4 - 7(x-2) &= 2(x-2)^2 \\ 4 - 7x + 14 &= 2x^2 - 8x + 8 \\ 2x^2 - x - 10 &= 0 \\ (2x-5)(x+2) &= 0 \\ \therefore x = \frac{5}{2}, x &= -2 \end{aligned}$$

10 b)

$$5^x = \frac{5}{2}, 5^x = -2$$

$5^x = -2$ has no solutions

$$x = \log_5\left(\frac{5}{2}\right)$$

11 a)

$$\begin{aligned} \log_2(x+3) &= \log_2 x^2 - \log_2 4 \\ \log_2(x+3) &= \log_2\left(\frac{x^2}{4}\right) \\ x + 3 &= \frac{x^2}{4} \\ 12 + 4x &= x^2 \\ x^2 - 4x - 12 &= 0 \\ (x-6)(x+2) &= 0 \\ \therefore x = 6, x &= -2 \end{aligned}$$

11 b)

$$\begin{aligned}
\log_4(5x+2) &= \log_4(13x-6) - 2\log_4 x \\
\log_4(5x+2) &= \log_4(13x-6) - \log_4(x^2) \\
\log_4(5x+2) &= \log_4\left(\frac{13x-6}{x^2}\right) \\
5x+2 &= \frac{13x-6}{x^2} \\
x^2(5x+2) &= 13x-6 \\
5x^3+2x^2 &= 13x-6 \\
5x^3+2x^2-13x+6 &= 0 \\
(x-1)(x+2)(5x-3) &= 0 \\
\therefore x = 1, x = -2, x &= \frac{3}{5}
\end{aligned}$$

$5 \times -2 + 1 < 0 \ \therefore$ reject $x = -2$

$$\therefore x = 1, x = \frac{3}{5}$$

11 c)

$$\begin{aligned}
-1 + \log_5(x^4+4) &= \log_5 x + \log_5(3x+14-5x^2) - \log_5 6 \\
\log_5(x^4+4) - \log_5 5 &= \log_5(x(3x+14-5x^2)) - \log_5 6 \\
\log_5\left(\frac{x^4+4}{5}\right) &= \log_5\left(\frac{3x^2+14x-5x^3}{6}\right) \\
\frac{x^4+4}{5} &= \frac{3x^2+14x-5x^3}{6} \\
6x^4+24 &= 15x^2+70x-25x^3 \\
6x^4+25x^3-15x^2-70x+24 &= 0 \\
(x+2)(3x-1)(x+4)(2x-3) &= 0 \\
\therefore x = -4, x = -2, x = \frac{1}{3}, x &= \frac{3}{2}
\end{aligned}$$

$3 \times -4 + 14 - 5(-4)^2 < 0$ so reject $x = -4$

$3 \times -2 + 14 - 5(-2)^2 < 0$ so reject $x = -2$

$$\therefore x = \frac{1}{3}, x = \frac{3}{2}$$

12)

$$\begin{aligned}
(\ln x)^2 + 4 &= \ln x^4 \\
(\ln x)^2 - \ln x^4 + 4 &= 0 \\
(\ln x)^2 - 4\ln x + 4 = (\ln x - 2)^2 &= 0 \\
\therefore \ln x &= 2 \\
x &= e^2
\end{aligned}$$

13)

$$\begin{aligned}
\sin^2 x &= \frac{7 - 13\sin x}{2} \\
2\sin^2 x &= 7 - 13\sin x \\
2\sin^2 x + 13\sin x - 7 &= 0 \\
(2\sin x - 1)(\sin x + 7) &= 0 \\
\therefore \sin x = \frac{1}{2}, \sin x &= -7
\end{aligned}$$

$\sin x = -7$ has no solutions

$\sin x = \frac{1}{2}$ has solutions $x = 30°, x = 150°$

14)

$$\begin{aligned}
\frac{\cos^2 x(3\cos x + 11)}{20} &= \cos x \\
\cos^2 x(3\cos x + 11) &= 20\cos x \\
3\cos^3 x + 11\cos^2 x &= 20\cos x \\
3\cos^3 x + 11\cos^2 x - 20\cos x &= 0 \\
\cos x(3\cos x - 4)(\cos x + 5) &= 0 \\
\therefore \cos x = -5, \cos x = 0, \cos x &= \frac{4}{3}
\end{aligned}$$

$\cos x = -5$ has no solutions

$\cos x = \frac{4}{3}$ has no solutions

$\cos x = 0$ has solutions $x = 90°, x = 270°$